6
Ethical Issues of Intelligent Automation Systems

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Abstract

Over the past decade, the growth of intelligent systems has tremendously increased in the working environment. It comprises cognitive technologies like artificial intelligence, robotic process automation, and business process management, which facilitate to decision-making process in various applications. In addition to that, it reduces operational costs, increases productivity, and improves accuracy with consistent processes for forecasting. Meanwhile, ethical considerations must be addressed to provide the sustainable and moral utilization of intelligent systems in any sector because the intelligent automation system (IAS) system must help the user to take an appropriate decision with more confidence. Hence, this chapter explores some of the major ethical issues in different domains and security problems that affect the process or execution of IAS and also discusses about the case studies on different ethical aspects of bias, deci-

sion-making on IAS, existing ethical frameworks, and environmental impacts of IAS in both positive and negative perspectives.

Keywords**:** IAS, ethical issues, bias, security, ethical frameworks, environmental impacts

6.1 Introduction

In the era of modern corporate, world organizations continually search for techniques that improve efficiency, simplify processes, and drive innovation. Intelligent automation systems, leveraging the power of artificial intelligence (AI), robotic process automation (RPA), and business process management (BPM), have emerged as indispensable tools for achieving these objectives. This article explores the fundamental concepts of AI, RPA, and BPM and examines how their integration into intelligent automation systems is revolutionizing the way businesses operate.

6.1.1 Artificial Intelligence

AI is a replica of natural human processes by computers. It encompasses various technologies like machine learning, natural language processing, and computer vision. AI enables machines to learn from data, acclimate to new inputs, and perform tasks that typically need human intelligence. In the business context, AI is being used for predictive analytics, customer service automation, and personalized marketing strategies [1].

In the ever-evolving landscape of technology, the concept of intelligence has taken center stage. From the advent of AI to the integration of smart systems in our daily lives, intelligence is not just a human attribute anymore; it is a characteristic embedded in the very fabric of our digital world. Among the myriad manifestations of intelligent design, the convergence of cutting-edge technologies has sparked a paradigm shift, redefining how we interact with machines, data, and each other.

6.1.2 The Rise of Artificial Intelligence

AI is the cornerstone of intelligent design. It incorporates a wide range of techniques that enable machines to imitate cognitive functions such as learning, problem-solving, and decision-making. What sets AI apart is its capacity to analyze huge amounts of data, recognize patterns, and adapt its behavior accordingly. This capability has paved the way for pioneering applications across various domains, from healthcare and finance to transportation and entertainment.

One of the most transformative applications of AI is in autonomous vehicles. Companies like Tesla, Waymo, and Uber are leveraging AI algorithms to develop self-driving cars that can navigate complex environments safely and efficiently. By processing data from sensors, cameras, and radar systems in real time, these vehicles can perceive their surroundings and make split-second choices to avoid complications and follow traffic rules.

6.1.3 The Internet of Things (IoT) and Smart Systems

The Internet of Things (IoT) is another cornerstone of intelligent design, connecting routine objects with the Internet and enabling them to interconnect with each other. This interconnected system of devices, sensors, and actuators has given rise to smart systems that can automate tasks, optimize processes, and enhance user experiences.

Smart homes, equipped with the help of IoT systems like smart regulators, illumination systems, and safety cameras, exemplify the transformative power of intelligent design. These systems can absorb user preferences, fine-tune settings based on ecological conditions, and provide remote access and control through smartphones or voice commands. As a result, homeowners can enjoy greater comfort, convenience, and energy efficiency while reducing their environmental footprint.

6.1.4 The Future of Intelligent Design

Looking ahead, the future of intelligent design holds boundless possibilities. Advances in AI, robotics, quantum computing, and biotechnology promise to revolutionize every aspect of our lives, from healthcare and

education to transportation and entertainment. However, realizing this vision requires a concerted effort to connect the power of technology for better advancement while mitigating its potential risks and drawbacks.

Ultimately, intelligent design is not just about creating smarter machines; it is about empowering individuals, enhancing human capabilities, and building a more inclusive and sustainable future for all. By embracing the principles of innovation, collaboration, and ethical stewardship, we can harness the transformative potential of intelligent design to tackle some of the most demanding challenges facing humanity and create a world that is smarter, safer, and more equitable for generations to come.

6.1.5 Robotic Process Automation (RPA)

RPA involves the practice of software robots or "bots" to mechanize repetitive, rule-based jobs. These bots can simulate human actions, such as logging on to some user-specific applications, doing the actions programmed by the user like copying/pasting data and filling out forms, generating reports, providing meaningful insights, and help with decision-making. RPA allows societies to streamline their workflows, reduce errors, and free up human employees to emphasize more value-added activities. It is particularly useful in industries like finance, healthcare, and manufacturing [2]. RPA has gained significant traction in recent years as organizations seek to mechanize repetitive, rule-based tasks to improve efficiency and reduce operational costs. While RPA proposes numerous benefits, it also presents challenges that organizations must steer to maximize its effectiveness. This article examines the challenges and benefits of RPA, drawing insights from industry research and case studies.

6.1.5.1 Benefits of Robotic Process Automation (RPA)

1. Enhanced Efficiency: RPA enables organizations to automate mundane and time-consuming tasks, letting employees to focus on more deliberate activities. According to a study by Deloitte, RPA can reduce process times by up to 90%, leading to significant efficiency gains [3].
2. Cost Savings: By automating repetitive tasks, RPA helps organizations reduce labor costs and improve resource utilization. A report by

McKinsey & Company suggests that organizations can achieve cost savings of up to 20–40% through RPA implementation [4].

3. Increased Accuracy: RPA bots complete tasks with a high level of accuracy and consistency, reducing errors and decreasing rework. This results in improved data quality and compliance, particularly in industries with stringent regulatory requirements such as finance and healthcare.
4. Scalability: RPA allows administrations to measure automation efforts rapidly to accommodate fluctuations in workload and business demands. This scalability enables organizations to react quickly to varying market conditions and seize new opportunities.

6.1.5.2 Challenges of Robotic Process Automation (RPA)

1. Complexity of Processes: Not all procedures are suitable for automation, particularly those that comprise composite decision-making or require human judgment. Identifying and selecting the right processes for automation is a key challenge faced by organizations implementing RPA [5].
2. Integration with Legacy Systems: Several official doms operate with legacy systems that may not be compatible with RPA platforms, posing challenges for integration and interoperability. Legacy systems often require additional customization and configuration to enable seamless automation [6].
3. Change Management: RPA implementation requires organizational change and may disrupt existing workflows and roles. Conflict between employees and stakeholders can hinder adoption and impact the success of RPA initiatives. Effective change management approaches are essential to address resistance and ensure successful implementation [7].
4. Governance and Compliance: RPA introduces new risks related to data security, privacy, and regulatory compliance. Organizations must establish robust governance frameworks and controls to mitigate these risks and ensure compliance with applicable regulations [8].

6.1.6 Business Process Management

Business process management (BPM) is a restraint that focuses on optimizing and managing business processes to improve efficiency and achieve organizational goals. It involves identifying, modeling, analyzing, and optimizing processes to enhance performance and agility. BPM software provides tools for process modeling, automation, monitoring, and optimization, allowing organizations to streamline their operations and adapt to changing markets [9]. BPM has become integral to modern organizations' efforts to enhance efficiency, agility, and customer satisfaction. While BPM offers several benefits, moreover presents challenges that administrations must address to realize its full potential.

6.1.6.1 Benefits of Business Process Management (BPM)

1. Process Optimization: BPM enables organizations to identify inefficiencies, streamline workflows, and eliminate bottlenecks, resulting in improved operational efficiency and productivity [9].
2. Enhanced Agility: BPM allows organizations to acclimate quickly to fluctuating market conditions, customer inclinations, and governing requirements. By automating and optimizing processes, organizations can respond more effectively to dynamic business environments [10].
3. Improved Customer Experience: BPM focuses on delivering value to customers by aligning processes with their needs and expectations. By mapping customer journeys and optimizing touchpoints, organizations can enhance customer satisfaction and loyalty [11].
4. Transparency and Visibility: BPM provides organizations with insights into their processes, performance metrics, and KPIs. This transparency enables stakeholders to make informed decisions, identify opportunities for improvement, and monitor progress toward strategic objectives [12].

6.1.6.2 Challenges of Business Process Management (BPM)

1. Resistance to Change: Implementing BPM often requires changes to existing processes, technologies, and organizational culture. Resistance from employees and stakeholders can hinder adoption and impact the success of BPM initiatives [13].

2. Complexity of Integration: Many organizations operate with disparate systems and technologies that may not integrate seamlessly with BPM platforms. Achieving integration and interoperability can be challenging, particularly in large, decentralized organizations [14].
3. Governance and Compliance: BPM introduces new risks related to data security, privacy, and regulatory compliance. Organizations must establish robust governance frameworks and controls to mitigate these risks and ensure compliance with applicable regulations [11].
4. Measurement and Performance Management: Measuring the effectiveness of BPM initiatives and demonstrating their impact on organizational performance can be challenging. Organizations must define relevant metrics, establish baseline measurements, and track progress over time to assess the success of BPM efforts [15].

6.2 Intelligent Automation Systems

Intelligent automation systems integrate AI, RPA, and BPM to create end-to-end automation solutions. These systems leverage AI systems to analyze data, make decisions, and enhance processes. RPA bots automate repetitive tasks, while BPM tools provide the framework for managing and optimizing business processes. By combining these technologies, administrations can achieve better efficiency, agility, and innovation in their processes [16].

6.2.1 Benefits of Intelligent Automation Systems

Intelligent automation systems offer numerous benefits to organizations, including:

1. ***Increased Efficiency:*** By automating repetitive tasks and optimizing workflows, intelligent automation systems can considerably reduce handling times and increase overall efficiency.
2. ***Cost Savings:*** Automation helps organizations save expenses by reducing the need for manual labor, minimizing errors, and maximizing resource utilization.
3. ***Enhanced Accuracy:*** AI-driven automation ensures greater accuracy and consistency in jobs such as data entry, document processing, and

quality control.

4. ***Improved Customer Experience:*** By streamlining processes and reducing turnaround times, intelligent automation systems can enhance the customer experience and increase satisfaction.

Before diving into the ethical implications of IAS, we discussed some of the application domains of IAS then what are all the ethical considerations that they faced are discussed in the below sections.

6.2.2 Healthcare Systems

The integration of intelligent automation within healthcare systems marks a pivotal advancement in revolutionizing patient care delivery, administrative processes, and operational efficiency. As we conclude the implementation of this transformative initiative, it is evident that intelligent automation has ushered in a new era of healthcare innovation, characterized by improved patient outcomes, enhanced operational efficiency, and increased cost-effectiveness.

- **Enhanced Patient Care:** By automating routine administrative tasks such as appointment scheduling, billing, and medical record management, healthcare providers can redirect their focus toward delivering high-quality, personalized patient care. Intelligent automation streamlines workflows, reduces administrative burdens, and enhances the overall patient experience.
- **Improved Operational Efficiency:** The automation of repetitive and time-consuming tasks within healthcare systems optimizes resource utilization, minimizes manual errors, and accelerates process completion. This newfound efficiency translates into reduced wait times, increased throughput, and enhanced staff productivity, ultimately leading to cost savings and improved operational performance.
- **Enhanced Accuracy and Compliance:** Intelligent automation systems leverage advanced algorithms and data analytics to ensure the accuracy and integrity of medical records, billing processes, and regulatory compliance. By minimizing human error and maintaining strict adherence to industry regulations, healthcare organizations can mitigate risks, enhance patient safety, and maintain regulatory compliance.

- **Empowered Healthcare Professionals:** Rather than displacing healthcare professionals, intelligent automation augments their capabilities by providing valuable insights, mechanizing repetitive tasks, and enabling data-driven decision-making. Healthcare professionals can leverage automation technologies to focus on critical patient care activities, clinical decision-making, and interdisciplinary collaboration, thereby improving overall healthcare delivery.
- **Scalable and Adaptive Solutions:** Intelligent automation solutions in healthcare are designed to be scalable and adaptable to evolving patient needs, technological advancements, and regulatory requirements. By adopting flexible automation platforms, healthcare organizations can future-proof their systems, seamlessly integrate new technologies, and respond rapidly to changing healthcare landscapes.
- **Challenges and Recommendations:** While the implementation of intelligent automation in healthcare systems presents numerous benefits, it also poses certain challenges, including data security concerns, interoperability issues, and resistance to change among healthcare professionals. Addressing these challenges requires a concerted effort to invest in robust cybersecurity measures, foster interoperability standards, and provide comprehensive training and support to healthcare professionals transitioning to automated workflows.

6.2.3 Business Process Management

Implementing an intelligent automation system within business process management marks a significant evolution in organizational efficiency and agility. This convergence of technology and strategy reshapes how businesses operate, unlocking new levels of productivity, accuracy, and innovation.

- **Enhanced Efficiency:** Intelligent automation streamlines workflows, reducing manual intervention and accelerating processes. Tasks that once required hours or days can now be completed in minutes, releasing valuable time for employees to emphasize high-value activities.
- **Improved Accuracy and Compliance:** By automating routine tasks, intelligent systems minimize human error, ensuring consistency and

compliance with regulations. This not only mitigates risks but also enhances the overall quality of operations.

- **Real-Time Insights:** The integration of intelligent automation facilitates data-driven decision-making by providing real-time insights into business processes. This empowers stakeholders to identify trends, anticipate challenges, and capitalize on opportunities promptly.
- **Agility and Adaptability:** Intelligent automation enables businesses to adapt swiftly to changing market dynamics and customer demands. Through dynamic process orchestration, organizations can reconfigure workflows on the fly, fostering agility and resilience in a rapidly evolving landscape.
- **Employee Empowerment:** Contrary to concerns about job displacement, intelligent automation empowers employees by augmenting their capabilities. By automating repetitive tasks, employees can focus on strategic initiatives, creativity, and innovation, fostering a culture of continuous improvement.
- **Customer-Centricity:** By automating backend processes, organizations can redirect resources toward enhancing the customer experience. This customer-centric approach fosters loyalty, satisfaction, and differentiation in a competitive marketplace.
- **Scalability and Growth:** Intelligent automation lays the foundation for scalable growth by optimizing resource allocation and minimizing operational bottlenecks. As businesses expand, automation ensures seamless scalability without compromising efficiency or quality.
- **Continuous Evolution:** The journey toward intelligent automation is not static but rather iterative. Organizations must embrace a culture of continuous improvement, leveraging feedback loops and emerging technologies to refine processes and maximize value over time.

6.2.4 Prompt Engineering

The implementation of an intelligent automation system using prompt engineering represents a significant milestone in our organization's journey toward operational excellence and digital transformation. This comprehensive endeavor has involved meticulous planning, strategic execution, and collaborative efforts across departments to harness the full potential of automation technologies. As we conclude this initiative, this re-

port provides a summary of key findings, achievements, and recommendations for future endeavors.

- **Assessment and Analysi:**: Through a thorough assessment of existing business processes, we identified several opportunities for automation, focusing on tasks characterized by high volume, repetitiveness, and potential for error reduction.
- **Strategic Planning:** A well-defined roadmap was developed, outlining the implementation strategy, timeline, and resource requirements. Clear objectives and success criteria were established, ensuring alignment with overarching business goals.
- **Technology Selection:** After careful evaluation, suitable automation technologies, including robotic process automation (RPA), AI, and machine learning (ML), were selected based on their compatibility with our requirements and objectives.
- **Design and Development:** Automated workflows were meticulously designed and developed in collaboration with stakeholders and IT teams, ensuring functionality, scalability, and seamless integration with existing systems.
- **Testing and Validation:** Rigorous testing and validation procedures were conducted to verify the reliability, functionality, and performance of the automation solution, incorporating feedback from end-users to refine the system.
- **Deployment and Rollout:** The deployment strategy was executed smoothly, with comprehensive training and support provided to end-users to facilitate the adoption of automated processes.
- **Monitoring and Optimization:** Continuous monitoring of automated processes enabled us to identify areas for optimization and refinement, ensuring ongoing efficiency and alignment with business objectives.
- **Documentation and Knowledge Sharing:** Documentation of the automation solution, including design specifications and best practices, facilitated knowledge sharing and collaboration among team members, fostering a culture of continuous learning and improvement.
- **Recommendations for Future Initiatives:** Continue to prioritize automation initiatives based on their potential impact on business outcomes and feasibility of implementation. Foster ongoing collaboration

between business stakeholders and IT teams to identify and capitalize on emerging automation opportunities. Invest in training and upskilling programs to empower employees with the necessary skills and knowledge to leverage automation technologies effectively. Explore opportunities for further integration of advanced technologies, such as natural language processing (NLP) and predictive analytics, to enhance the capabilities of the automation system. Establish mechanisms for continuous monitoring, optimization, and refinement of automated processes to ensure alignment with evolving business needs and objectives.

While an intelligent automation system offers the above significant benefits, it also poses challenges, ethical issues, and considerations for organizations, which are discussed in Section 6.3.

6.3 The Ethical Implications of Intelligent Automation Systems

While intelligent design holds immense promise for improving our lives and advancing society, it also raises significant ethical concerns. Chief among these is the issue of job displacement, as automation and AI-driven technologies replace human workers in various industries. Additionally, there are concerns about data privacy, security, and the potential misuse of AI for surveillance, manipulation, or discrimination.

To address these challenges, policymakers, technologists, and ethicists must work together to develop robust regulations, guidelines, and ethical frameworks for the responsible design and deployment of intelligent systems. This includes ensuring transparency and accountability in AI algorithms, protecting user data and privacy rights, and promoting inclusivity and diversity in technology development. This section discusses the following ethical issues of IAS:

1. ***Data Security and Privacy***
2. ***Lack of IT Readiness & Skill Gap***
3. ***Decision-Making Process of IAS***
4. ***Data Sharing Policies and Regulations***

5. ***Bias and Fairness in IAS***

6.3.1 Data Security and Privacy

In this digital era, almost in all domains processing of data becomes an indispensable thing. At the same time how we treat and store them is also very essential, especially when dealing with sensitive information such as medical records, user's personal data, or company information. Therefore, keeping them safe and availing permission to authorized persons are also important. It involves seeking for permission, reporting to others what we have done, and abiding by the rules about how we can use the information of another security system. Protecting personal data from unauthorized access and ensuring that the data is not modified by the same are referred to as data privacy and security, respectively. Keeping information out of unauthorized hands is the goal of data security. It includes techniques to guarantee that data is not transferred, changed, or damaged [9, 14]. Companies need to take precautions to protect sensitive data and make sure that data protection laws are followed. Using privacy-preserving methods, including encryption and anonymization, is essential to stop unwanted access and safeguard the identities of people whose data AI systems utilize [17].

6.3.1.1 Security Problems

Needs Constant Supervision

To make sure intelligent systems are operating correctly and securely, constant monitoring is necessary, especially for those that use machine learning. In the absence of consistent supervision, these systems may exhibit prejudices, condense incorrect judgments, or expose themselves to security risks. Frequent oversight guarantees that vulnerabilities are addressed, updates are applied to the systems, and the models are retrained using pertinent data [18].

Great Exposure

Intelligent systems are very vulnerable to possible threats because of their deep integration with an extensive selection of networks and data

sources. Because of their high visibility, they are desirable targets for cybercriminals who can use loopholes to obtain private data without authorization [19].

Data Abuse

The misuse of information gathered by intelligent systems is referred to as data abuse. Unauthorized access, careless handling, or deliberate insider misuse can all lead to this. Preventing data abuse requires ensuring data privacy and protection [20].

- ***Unauthorized Access to Personal Data:*** Sensitive personal health data may be stored by an intelligent system, such as a health monitoring app. Hackers may obtain unauthorized access to this data if this system is not sufficiently secured, which could result in privacy violations and the misuse of sensitive health information. Hackers gained access to 78.8 million people's medical records from Anthem Inc., a significant health insurance provider, in 2015. Personal data, including names, Social Security numbers, and medical records, were compromised [21].
- ***Surveillance and Tracking:*** Intelligent systems that have facial recognition software installed can be used to follow and monitor people without permission. This may result in serious privacy problems and improper usage of personal information. In 2019, a Chinese corporation was tracking the activities of the Uighur Muslim minority in China using face recognition technology, and major concerns over invasions of privacy and human rights were raised [22].
- ***Misuse of Social Media Data:*** Social media networks gather and examine user data through intelligent systems. Misinformation campaigns and targeted political advertising are just two uses for this data that could be abused. The 2018 Cambridge Analytica scandal concerned the improper use of millions of Facebook users' personal information. There are serious ethical and privacy issues because the data was used to build comprehensive voter profiles and sway election results [23].
- ***Selling User Data Without Consent:*** Organizations that create intelligent systems run the risk of collecting and selling user data without

permission, which is against ethical and privacy guidelines. It came to light in 2019 that a number of well-known health applications were improperly sharing user data with Facebook and other third-party businesses. Sensitive health information and usage trends were included in this report [24].

Unsecure Frameworks

A lot of intelligent systems may be constructed using non-secure frameworks. These frameworks may contain security holes that hackers could take advantage of if they are not fixed. It is crucial to guarantee the security of these frameworks by conducting frequent updates and security audits [25].

Outlays of Non-compliance

There can be severe financial fines and reputational damage for breaking rules and norms. To prevent these expenses, intelligent systems have to go by a number of compliance mandates, including GDPR, HIPAA, and other industry-specific laws [26].

Unproductive Bots

Bots are frequently used by intelligent systems for a variety of functions. Unproductive bots, on the other hand, might result in resource waste and possible security flaws if they fail to carry out their intended tasks properly or efficiently. To guarantee efficiency and security, these bots need to be regularly assessed and optimized [27].

Audit Long Maintenance

Prolonged maintenance intervals amid audits may result in system outages and greater vulnerability. In order to reduce downtime and guarantee uninterrupted security and compliance, audit procedures must be streamlined and efficient [28].



Figure 6.1 Adversarial attack types of intelligent systems.

The growth and use of IAS systems are widespread, at the same time, while considering security concerns, they are vulnerable to adversarial attacks, which are illustrated in Figure 6.1. In [29], the author classifies some of the adversarial attacks which affect the performance of intelligent systems.

6.3.2 Lack of IT Readiness & Skill Gap

Many organizations still mostly rely on centralized IT operations to execute intelligent automation because they are overworked and unable to handle the demands of maintaining business data, systems, and information security. IAS systems handle huge volumes of data, which may include structured and unstructured data. To process them, it requires

much more computing and storage and other infrastructure resources, which are on the ground, and it must rely on cloud storage for scalability and capacity reasons. Moreover, it also requires experienced person who already have knowledge of working with these cloud storages.

IAS results in a dramatic shift in operational procedures, and because of advances in cognitive computing technology, the great popular cognitive functions can now be mechanized with no human participation [29]. Management should support staff members in seeing IAS as a digital worker and make sure they have enough time, resources, and training to use IAS properly. The familiarity or working procedure of the system must be known to the end user of the system unless it would be the cause of some other problems. Aside from that, management needs to understand that implementing IAS involves reengineering human resources rather than replacing employees [11].

- To conquer a lack of IT readiness IT representation must be integrated in early and actively in all IAS projects. Also, the data and integrations must be maintained in accordance with important organizational principles.
- Leaders in automation must also comprehend the IT viewpoint. When new technologies are implemented, a variety of enterprise-wide technology projects, infrastructure, and IT capacity and skills may be impacted; these matters need to be taken into account.

6.3.3 Decision-Making by IAS

IAS has become an influential trend that is changing how decisions are made in industries. This technological growth simulates the cognitive processes in humans to improve results and efficiency. Determining who is accountable for the judgments made by intelligent automation systems is one of the most critical issues in AI ethics. The boundaries between responsibilities become more unclear when these systems carry out more complex operations, such as financial investment decisions and illness diagnoses.

Fundamentally, IAS is the creation of intelligent agents that are able to sense their surroundings, process information, and make judgments in order to accomplish particular objectives [12, 13]. Transparency in IAS makes it accessible and intelligible to those who will be adversely affected by the results. People must understand the process by which AI algorithms make particular decisions [14].

6.3.4 Data Sharing Policies and Regulations

Policies, regulations, and standards regarding data sharing for intelligent systems differ greatly based on the industry, jurisdiction, and kind of data being managed. Some of the existing data-sharing policies are listed in Table 6 1.

Data Privacy Laws: Robust data privacy laws that control the collection, use, and sharing of personal data have been passed by several nations. The California Consumer Privacy Act (CCPA) in the United States and the General Data Protection Regulation (GDPR) in the European Union are two examples. It introduces incentive models for large companies to share their aggregated data, which helps for faster economic growth of IAS [30].

Sector-Specific Regulations: Sometimes the information that we stored was sensitive, so they have to be kept in safe with more accessing protocols. This type of data may be shared with several industries like banking and healthcare. The Health Insurance Portability and Accountability Act (HIPAA) in the United States is an example of overseeing the exchange of medical information in the healthcare industry [31].

Data Proprietorship: Intelligent systems frequently work with data that originates from or is gathered from different sources. To establish who can share the data and under what circumstances, explicit policies pertaining to data ownership and intellectual property rights are required.

Table 6.1 Data sharing policies.

Policy/regulation	Year	Key features
Health Insurance Portability and Accountability Act **(HIPAA)** - United States [32]	**1996**	• Requires patient consent for data sharing. • Establishes national standards for electronic health care transactions. • Charge penalties for violations.
Personal Data Protection Act **(PDPA)** - Singapore [33]	**2012**	• Requires organizations to obtain consent before collecting, using, or disclosing personal data. • Allows individuals to access and correct their personal data. • Imposes fines for non-compliance.

Policy/regulation	Year	Key features
General Data Protection Regulation **(GDPR)** - European Union [34]	**2018**	• Needs explicit consent for data collection and sharing. • Offers individuals with the right to access, rectify, and delete their data. • Orders data breach notifications within 72 hours. • Enforces severe penalties for non-compliance.
California Consumer Privacy Act **(CCPA)** - United States [35]	**2020**	• Enables customers to have an opinion in what personal information is gathered and how it is used. • Enables customers to choose not to have their personal data sold. • Charges fines for failure to comply.

Informed Consent: People have the right to agree to the collection and use of their personal data in many different jurisdictions. This needs open communication regarding the usage and sharing of data, as well as giving people the option to choose out if they so desire.

Anonymization and De-Identification: Data shared by intelligent systems may need to be removed from identification or anonymized before sharing in order to safeguard privacy by removing personally identifiable information. A great deal of caution is required because even

anonymized data can occasionally be re-identified using a variety of methods.

Cross-Border Data Transfers: There can be further rules in place when sharing data from one country to another, in other words internationally. Certain legal frameworks prohibit the export of data to nations that do not offer sufficient safeguards for personal information.

Ethical Considerations: The ethical implications of data sharing in intelligent systems are significant in addition to legal responsibilities. Making decisions should be based on values like equity, responsibility, openness, and respect for the rights of persons.

Regulatory Compliance: In order to prevent fines and harm to their reputation, companies that create and implement intelligent systems must keep up with changing regulatory requirements and make sure that all applicable laws and standards are followed.

6.3.5 Bias and Fairness in IAS

The explanation for AI biases is that data can be transferred from humans to AI through the programming and coding process, which leads to the development of the racism and discrimination problem. In other words, it defines that bias is an anomaly caused by machine learning algorithms during the development of the algorithm or in the selection of training datasets. If the given input is biased then obviously the result given by the intelligent system is also biased. The problem of bias in AI systems is complex and can originate from different phases of the AI development phases, such as data gathering, algorithm creation, model training, and implementation.

Data Collection

- ***Historical Bias:*** When historical prejudice or inequalities are reflected in the data used to train AI models, it is known as historical bias. For example, if recruiting procedures in the past have been discriminatory against women, an AI recruitment system trained on previous hiring data may learn to favor male candidates.

- ***Sampling Bias:*** It occurs when the AI is not trained on data that is representative of the population it will serve. For example, when applied to darker-skinned people, a facial recognition system that was predominantly trained on photos of light-skinned people may not function well.

Algorithm Design

- ***Objective Function Bias:*** An AI model's objective function, which specifies what the algorithm seeks to maximize, has the potential to generate bias if fairness is addressed. Example: If minority groups' financial behavior patterns diverge from those of the majority, a credit score system that just considers profit may disproportionately hurt them.
- ***Feature Selection Bias:*** When choosing features for the model, bias may be introduced if those features have a correlation with sensitivity factors like gender or ethnicity. For example, ZIP codes frequently correspond with socioeconomic and ethnic characteristics, therefore including them as features in a loan acceptance model may introduce these biases.

Model Training

- ***Training Data Imbalance:*** When applied to groups who are underrepresented the model may perform badly due to imbalanced training data. For example, An AI healthcare system may not correctly diagnose problems in female patients if it is trained mostly on data from male patients.
- ***Overfitting to Biased Data:*** A model is said to be overfitted when it learns biases and other noise from training data instead of the real underlying patterns. Example: Existing prejudices in the legal system may be perpetuated if an AI model trained on skewed court data learns to anticipate harsher sentences for particular demographic groups.

Deployment

Feedback Loops: *When an AI system's outputs confirm the biases found in the input data, an endless cycle of bias propagation is created. This is known as a feedback loop. Predictive policing that disproportionately impacts communities inhabited by minorities may increase arrests in such areas, so validating the biased data and predictions.*

Contextual Bias: *When an AI system is used in an environment other than the one it was trained for, bias may be introduced. For example, in customer assistance for clients who do not speak English or come from diverse cultural backgrounds, a Chabot that was trained on English-speaking customers in the US might not function properly.*

6.3.5.1 Case Studies for Bias in AI Systems

i) Amazon's AI Recruiting Tool: Amazon created a recruiting tool with AI to automate the evaluation of job applications and simplify the hiring process. The tool's purpose was to assess resumes and assign a candidate's chance of success in different areas within the organization. However, because of the makeup of the training data it was given, it was discovered that the tool was biased against women [36].

Objective of the system: *The tool was proposed to help hiring managers find out the best candidates by analyzing and scoring resumes.*

Training Phase: *The program was trained on resumes over a ten-year period, mostly from male candidates, representing the time when the tech industry was dominated by men.*

Bias Identification: *The algorithm consistently ranked resumes with terms associated with women lower, preferring language and experience characteristic of professions controlled by men, such as all-women institutions or women's groups.*

The Implications and Effects: *Due to the bias, female applicants were discriminated against, and competent individuals may have been disqualified based on unfair criteria. The incident brought to light the larger problem of prejudice in AI systems.*

Response and Resolution: *After discovering the bias, Amazon stopped using the AI recruiting tool. The event made clear how important it is to thoroughly test and assess AI systems in order to identify and reduce biases before implementation.*

ii) Google Photos: Due to biased training data, it was discovered that Google Photos, an image storage and organization service, incorrectly identified photos of African Americans as "gorillas" [37].

Objective of the System: *Google Photos automatically labels and categorizes photos using machine learning algorithms, which facilitates users' ability to search and manage their photo collections.*

Training Phase: *The dataset used to train the algorithm was not sufficiently diverse, especially when it came to pictures of persons with darker skin tones.*

Bias Identification: *In 2015, a software engineer found that Google Photos had labeled photos of his African American friends as "gorillas," highlighting an important issue with the algorithm's capability to correctly recognize darker skin tones.*

The Implications and Effects: *The incident brought attention to the possibility of racial biases in AI systems and incited public concern. In order to stop such offensive errors, it underlined the necessity for more inclusive and representative training datasets.*

Response and Resolution: *Google expressed an apology for the event and worked quickly to resolve the problem. In the process of working on longterm algorithmic enhancements, the corporation temporarily deleted the "gorilla" label from its image recognition system.*

As a result, bias may find its way into algorithms, which are only useful for systems' training datasets. Bias and fairness are among the most significant factors to consider in the deployment of IAS. Without addressing these, it may hinder the firm's progress and lead to unintended consequences. For example, gender bias will affect one part of a group which leads them to production loss. Likewise, it should not be biased in all the scales. The trained data must be able to be capable of carrying out these things to make the IAS system give an appropriate response to attain the objective goals of the system. Table 6.2 describes some of the algorithmic bias issues in popular firms.

Table 6.2 Some of the noticeable algorithmic bias.

Company	Algorithmic bias
Adobe	Customers belonging to particular demographics were prevented from purchasing Adobe software [38].
Facebook	Facebook allows advertisers to tailor their job and marketing advertising to specific demographics, including gender, color, and religion [39].
Banker	Payment processing errors and revenue-sharing agreements [40].
Microsoft	When users began interacting with the chatbot about racist remarks, the voice assistant began to type the same sentences again rather than responding to the user's questions [41].
Pulse oximeter	It is one of the key new tools for clinical care during pandemics used as oxygen level monitoring gadget. It prejudice that darker skin types receive fewer accurate results than lighter skin types (racial bias) [42].

6.4 Case Studies of Ethical Issues in IAS Decision-Making

IAS brings efficiency and innovation to many different industries, but alongside the benefits in many areas, there are also crucial ethical issues such as accountability and responsibility, transparency and explicable, and bias and fairness. In addition, there are four ethical principles and their corresponding challenges that are mentioned here.

- **Autonomy:** The patient refuses effective treatment due to personal beliefs.

- **Beneficence:** Access to optimal treatment is limited by a lack of resources.
- **Justice:** The patient chooses a riskier therapy that may be less effective.
- **Non-malfeasance:** Insufficient knowledge of risks and treatment alternatives.

Here, we illustrate some case studies that state the incidents focus on the complex interaction between technological advancements and moral principles. For example, Amazon's recruitment tool which was developed to assess the resumes was found that it exhibits gender bias. Because the algorithm was trained on recruiting data from the past, it reflected the biases present in that data. This instance emphasizes the moral difficulties in applying AI in situations where past data could reinforce or magnify already-existing disparities. The transparency of situations such as the Amazon hiring tool emphasizes how crucial it is to comprehend how algorithms work [38, 43].

6.4.1 Ethical Issues in Healthcare

It is important to consider the need to follow the conventional principles of medical ethics, familiarize old notions with new procedures of communication, and discourse the distinctive ethical concerns one may come upon with the use of technology in healthcare [44]. Ethical questions raised by emerging mental health-based chatbots [45]. To simplify patient triage in the emergency room, a major hospital uses an IAS. Let us dive into this. Prioritizing patients according to the severity of their conditions and the anticipated length of time they would have to wait for treatment, the IAS employs AI algorithms to examine patient symptoms, medical histories, vital signs, and other data. Let us examine the breast cancer detection and remote diabetes management, listed in Table 6.3.

Table 6.3 IAS healthcare ethical consideration.

Tools	Technology	Benefits	Ethical considerations
Cancer detection	Deep learning for mammogram analysis	Improved accuracy, reduced workload, personalized risk assessment	Bias in training data, over-reliance on AI
Remote diabetes management	AI-powered chatbots	Improved selfmanagement, enhanced adherence, reduced costs, increased accessibility	Data privacy & security, digital divide, dependence on technology

• ***Cancer Detection:***

The use of deep learning for mammogram analysis brings significant benefits, including enhanced accuracy in breast cancer detection, reduced workload for radiologists, and personalized risk assessment for patients. However, several ethical considerations must be addressed. Firstly, biases in training data may lead to inaccuracies, particularly affecting underrepresented groups. Secondly, there is a concern about over-reliance on AI, which could diminish critical thinking among healthcare professionals and result in a failure to verify AI-generated results and analyse the risks and benefits of cancer detection technology. These ethical challenges highlight the importance of ensuring fairness, transparency, and accountability in the implementation of deep learning technology for mammogram analysis to maximize its benefits while minimizing potential harms [46].

• ***Remote Management:***

Remote management using AI-powered chatbots offers numerous benefits, including improved self-management, enhanced treatment adherence, reduced healthcare costs, and increased accessibility to healthcare services. However, several ethical considerations must be addressed to ensure responsible implementation. Firstly, protecting patient data privacy and security is paramount to maintaining confidentiality and complying with regulations. Secondly, the digital divide poses a challenge, as not all patients may have access to the technology required to utilize AI-powered chatbots, potentially widening healthcare disparities. Lastly, there is a risk of dependence on technology, which could diminish the significance of human interaction and clinical judgment in diabetes management. Addressing these ethical considerations is crucial to harnessing the full potential of remote diabetes management while prioritizing patient well-being and equity in healthcare delivery [47]. Implement an ethical evaluation methodology to assist in detecting significant ethical considerations in future Digital Disease Detection studies.

Table 6.4 IAS ethical consideration in NLP-based tools.

Application	How it works	Benefits	Ethical considerations
Automating fraud detection in transactions	NLP analyzes transaction details like text descriptions and customer information to detect patterns indicating potential fraud, reducing manual workload and financial losses.	Reduces manual workload Improves accuracy and speed of fraud detection Minimizes financial losses	*False positives*: Inconvenience caused by overly sensitive algorithms flagging legitimate transactions *Bias:* Unfair targeting of certain demographics due to biases in training data
Analyzing market trends from news reports	NLP analyzes news articles and financial reports to identify emerging trends and market sentiment, providing insights for informed investment decisions.	• - Provides insights into market sentiment • - Identifies potential risks and opportunities • - Supports data-dri-	*Misinformation*: Challenges with sarcasm or manipulation in news reports leading to skewed predictions *Limited Scope:* Potential for missing crucial information not explicitly stated in text

Application	How it works	Benefits	Ethical considerations
		ven in-vestment strate-gies	
Personalizing financial advice based on customer data	NLP analyzes customer financial history and preferences to offer tailored financial guidance and product recommendations, improving customer satisfaction and commitment.	Offers tailored financial guidance Improves customer satisfaction Increases engagement with financial products	*Bias*: Unfair recommendations due to biases in NLP model or customer data, exacerbating financial inequality *Transparency*: Need for updating customers about data usage and provided that opt-out options
Analyze customer reviews, emails, social media posts, etc. to understand sentiment and extract themes.	NLP algorithms trained on labeled text data identify sentiment based on language cues.	Gain customer satisfaction insights, improve customer service, track brand perception, personalize marketing.	NLP may struggle with sarcasm, humor, or cultural references, leading to misclassified sentiment.

6.4.2 Ethical Considerations in NLP-Based Tools

NLP tools are having a transformative impact across numerous applications such as customer service, education, finance, and content creation, as given in Table 6.4.

6.4.3 Ethical Considerations in Prompt Engineering

Prompt engineering refers to strategically designing effective prompts or input instructions to shape the responses generated by AI language models. It is becoming a popular subfield in AI and natural language processing. Prompt engineering helps to create and optimize inputs for using language models efficiently, i.e., Double-Edged Sword of Prompt Engineering.

Numerous privacy ethics issues persist despite advancements in technology. EHR concentrated qualitative research on ethics, archives, records management papers, and patient and healthcare provider communications [48]. Tele-IAS (telemedicine intelligent automation systems) are revolutionizing remote healthcare by leveraging the power of AI and machine learning [49]. Electronic health record (HER) offers a wealth of data for research but raise new ethical concerns [50]. Table 6.5 shows the summary of ethical considerations in respect of prompt engineering-based IAS.

Table 6.5 IAS prompt engineering tools ethical consideration.

Tools	Goal	Approach	Results	Ethical considerations
Improving customer sentiment analysis	More accurate sentiment analysis of customer reviews	Craft prompts to consider sarcasm & language nuances, use context from surrounding text	Increased accuracy in sentiment analysis, better customer insights	Transparency about AI use, human oversight for complex cases
Generating creative content	Create engaging & informative blog posts	Define content style & target audience, craft prompts with structure & creative inspiration	Enhanced content creativity & efficiency	Transparency about AI-generated content, prevent plagiarism

6.4.4 Job Displacement

The rise of automation sparks concerns about job displacement, leading to economic strain [51] and social instability due to increased unemployment. This trend can exacerbate existing inequalities, disproportionately affecting certain demographics. To address this, there is a pressing need for continuous upskilling, yet access to retraining programs and education remains unequal.

6.5 Environmental Impacts of IAS

AI, machine learning (ML), robotics, and the Internet of Things (IoT) are just a few of the technologies that are being included in intelligent automation systems (IAS) and are revolutionizing various sectors and societies. These systems have a big influence on the environment even if they have several advantages. Here, some of the positive environmental impacts and the key factors that affect the environment are listed in Table 6.6 and Table 6.7.

Table 6.6 Positive environmental impacts of IAS.

Component	Description
Energy efficiency and resource management	AI algorithms in smart grids can optimize energy distribution, minimizing waste and boosting efficiency. Accurately forecasting energy demand allows power plants to limit excess production, thereby reducing greenhouse gas emissions [53]. According to a McKinsey & Company report from 2020, demand response systems powered by AI in smart grids can lower energy consumption by as much as 10% [52].
Waste reduction	Optimizing the utilization of raw materials and enhancing production precision through intelligent automation can reduce material waste in manufacturing processes. The World Economic Forum claims that the usage of AI in manufacturing might result in up to a 20% reduction in waste [54].
Sustainable agriculture	Precision farming makes better use of water, fertilizer, and pesticides by using AI to monitor crop health, soil conditions, and weather patterns. According to the European Commission, precision farming can boost crop yields by 10% to 15% while using 30% less water [55].
Wildlife conservation	Drones with AI and camera traps can track poachers and keep an eye on animal populations, supporting conservation efforts. Through the application of AI, a World Species Fund project is able to better monitor and

Component	Description
	protect species by analyzing data from video traps [56].

Table 6.7 Negative Environmental Impacts of IAS.

Component	Description
Energy consumption	Large amounts of electricity are needed for the operation and cooling of data centers, which house the servers and infrastructure for AI and IoT applications. It is estimated that data centers accounted for about 1% of global electricity demand in 2020. Training large AI models, such as deep neural networks, requires significant computational power and energy. Over the span of their training, one AI model can produce an equal amount of carbon as five cars [57].
E-Waste	AI and automation technologies are developing rapidly, outdated equipment must be discarded and upgraded frequently, which adds to the growing amount of electronic trash (e-waste). The global capacity of e-waste generated in 2019 was nearly 53.6 million metric tons [58].
Resource depletion	The extraction of electronic systems for manufacturing electronic devices that use rare earth metals has substantial environmental impacts. The usage of lithium batteries can be the cause of soil degradation and pollution [59].

6.6 Existing Ethical Frameworks of IAS

IAS ethical frameworks seek to guarantee that these systems are established, implemented, and used in a way that maximizes benefits to society and minimizes harm. The following are a few of the current ethical frameworks:

- **Asilomar AI Principles:** Guidelines for AI research and improvement are provided by the Future of Life Institute's Asilomar AI Principles. These guidelines place a strong emphasis on being transparent, safety, and the ethical integration of AI [60].
- **IEEE Global Initiative on Ethics of Autonomous and Intelligent Systems:** The goal of the IEEE project is to make sure that moral issues are taken into account when creating intelligent and autonomous systems. The "Ethically Aligned Design" document outlines several important ethical principles which are human rights, accountability, and the well-being of society and individuals [61].
- **European Commission's Ethics Guidelines for Trustworthy AI:** The European Commission has created criteria to guarantee the reliability of AI systems. Three key pillars serve as the foundation for these guidelines: ethical, robust, and legal AI [62].
- **UK House of Lords AI Code:** The UK House of Lords has proposed a code for AI that includes five principles to guide AI development and deployment.
- AI development ought to be done with humanity's best interests in mind.
- AI should function according to fairness and clearness standards.
- AI should never be used to violate a person, family, or community's privacy rights or data rights.
- Every citizen has the right to an education that will help them coexist peacefully with AI and thrive on all mental, emotional, and financial levels.
- AI should never be given the independent ability to harm, destroy, or trick people [63].

6.7 Conclusion

Intelligent automation systems have become indispensable tools in now and the future. The deployment of IAS with the integration of AI, RPA, and BPM makes the system more productive, decreases operational costs, and guarantees improved accuracy. Though with these attainments, it should be concentrating on the ethical issues to ensure the responsible and ethical use of IAS, which are discussed in this chapter as data privacy and security play vital roles that protect data from unauthorized users since the data they handle may be medical data, personal data, or some other sensitive data. Then, the lack of IT readiness, which is the IAS system, handles the voluminous amount of data so it should be scalable with the required infrastructure. Meanwhile, the user must also have the proper knowledge about the system because the skill gap will also lead to erroneous situations. Sometimes the decision-making by the IAS systems is also evaluated with the expert's judgments to mark the system's transparency.

Another important issue is the IAS system must follow the standard data-sharing regulation policies, which incorporates the following components as data privacy laws, sectorspecific regulation, data proprietorship, ethical consideration, regularity compliance, etc. Further, the bias and fairness of the system will make the smooth behavior of the production in a positive direction if it fails it throwbacks. The development or deployment of the IAS system must be taking all these issues into consideration. The following are some of the potential solutions for the above issues: diverse training data, human oversight and explainable AI, regulation and ethical guidelines, transparency and open communication, and human-robot collaboration for use and advancement in the field of intelligent automation systems in all sectors.

References

1. Duggan, M., Robotic Process Automation, 2020, Retrieved from https://www.gartner.com/en/information-technology/glossary/robotic-process-automation-rpa.

2. Lacity, M.C. and Willcocks, L.P., Robotic process automation at Xchanging. *Strategic Outsourcing: Int. J.*, 9, 3, 255–272, 2016.
3. Russell, S.J. and Norvig, P., *Artificial Intelligence: A Modern Approach*, Pearson, 2022. https://people.engr.tamu.edu/guni/csce421/files/AI_Russell_Norvig.pdf
4. Weske, M., *Business Process Management: Concepts, Languages, Architectures*, Springer, 2019. https://www.hira-bpm.ir/wp-content/uploads/2023/05/Business-Process-Management_-Concepts-Languages-Architectures-Springer-Berlin-Heidelberg-2019.pdf
5. Deloitte, RPA in the Age of Hyperautomation, 2020, Retrieved from https://www2.deloitte.com/us/en/insights/focus/cognitive-technologies/robotic-process-automation.html.
6. KPMG, RPA Governance, 2019, Retrieved from https://advisory.kpmg.us/content/dam/advisory/en/pdfs/2019/rpa-governance.pdf.
7. Marr, B., Why Robotic Process Automation Is The Future Of Shared Services, 2018, Retrieved from https://www.forbes.com/sites/bernardmarr/2018/07/09/why-robotic-process-automation-is-the-future-of-shared-services/?sh=22162a08458a.
8. McKinsey & Company, The Next-Generation Operating Model for the Digital World, 2017, Retrieved from https://www.mckinsey.com/business-functions/operations/our-insights/the-next-generation-operating-model-for-the-digital-world.
9. Nelson, R. and Moody, G., RPA Change Management: Building the Foundation for Automation Success, 2019, Retrieved from https://www.isg-one.com/research/articles/rpa-change-management-building-the-foundation-for-automation-success.
10. Peppard, J. and Ward, J., *The strategic management of information systems: Building a digital strategy*, John Wiley & Sons, 2016. https://www.wiley.com/enit/The+Strategic+Management+of+Information+Systems%3A+Building+a+Digital+Strategy%2C+4th+Edition-p-9780470034675.
11. Gartner, Enhancing the Customer Experience With Business Process Management, 2020, Retrieved from https://www.gartner.com/en/information-technology/insights/business-process-management.
12. Gartner, Governance, Risk and Compliance in Business Process Management, 2021, Retrieved from https://www.gartner.com/en/infor-

mation-technology/insights/business-process-management-governance-risk-and-compliance.

13. Harrington, H.J. and Voehl, F., *Business Process Improvement Workbook: Documentation, Analysis, Design, and Management of Business Process Improvement*, McGraw-Hill Education, 2013. https://cdn.chools.in/LEAN_PDF/Business-ProcessImprovement,Dr.%20H.%20J.%20Harrington.pdf.
14. Harmon, P., *Business Process Change: A Guide for Business Managers and BPM and Six Sigma Professionals*, Morgan Kaufmann, 2015. https://www.researchgate.net/publication/49110422_Business_Process_Change_A_Guide_For_Business_Managers_and_BPM_and_Six_Sigma_Professionals_P_Harmon_prol_de_TH_Davenport.
15. Hevner, A.R., March, S.T., Park, J., Ram, S., Design Science in Information Systems Research. *MIS Q.*, 28, 1, 75–105, 2004.
16. Jeston, J. and Nelis, J., *Business Process Management: Practical Guidelines to Successful Implementations*, Routledge, 2014. https://www.taylorfrancis.com/books/mono/10.4324/9780203081327/business-process-management-john-jeston.
17. Pierce, S. and Swatman, P.M., *Business Process Management and the Balanced Scorecard: Using Processes as Strategic Drivers*, Routledge, 2014. http://ndl.ethernet.edu.et/bitstream/123456789/29239/1/46.Ralph%20F.%20Smith.pdf.
18. Mahmoud, R., Yousuf, T., Aloul, F., Zualkernan, I., Internet of Things (IoT) security: Current status, challenges, and prospective measures, in: *10th International Conference for Internet Technology and Secured Transactions (ICITST)*, IEEE, pp. 336–341, 2015.
19. Roman, R., Zhou, J., Lopez, J., On the features and challenges of security and privacy in distributed internet of things. *Comput. Netw.*, 57, 10, 2266–2279, 2013.
20. Sweeney, L., k-Anonymity: A model for protecting privacy. *Int. J. Uncertainty Fuzziness Knowledge Based Syst.*, 10, 05, 557–570, 2002.
21. Abelson, R. and Creswell, J., Millions of Anthem Customers Targeted in Cyberattack, The New York Times, 2015, February 5.
22. Mozur, P., One Month, 500,000 Face Scans: How China Is Using A.I. @ to Profile a Minority, The New York Times, 2019, April 14.
23. Rosenberg, M., Confessore, N., Cadwalladr, C., How Trump Consultants Exploited the Facebook Data of Millions, The New York

Times, 2018, March 17.

24. Timberg, C. and Harwell, D., Health apps share your concerns with advertisers, The Washington Post, 2019, February 22.
25. Xie, X., Lu, Y., Ding, W., Cheng, Z., Security risks and countermeasures in cloud computing. *J. Netw.*, 9, 1, 166–170, 2014.
26. Pwc, Global Data Protection and Privacy Law, 2020.
27. Ferrara, E., Varol, O., Davis, C., Menczer, F., Flammini, A., The rise of social bots. *Commun. ACM*, 59, 7, 96–104, 2016.
28. Böhme, R. and Moore, T., The economics of cybersecurity: Principles and policy options. *Int. J. Crit. Infrastruct. Prot.*, 4, 3-4, 132–134, 2012.
29. R. S. Lee AI Ethics, Security and Privacy, in: *Artificial Intelligence in Daily Life*, 2020. DOI: 10.1007/978-981-15-7695-9_14.
30. https://www.consilium.europa.eu/en/policies/data-protection/data-protection-regulation/
31. https://www.hhs.gov/hipaa/for-professionals/compliance-enforcement/index.html
32. McGraw, D., Building public trust in uses of Health Insurance Portability and Accountability Act de-identified data. *J. Am. Med. Inf. Assoc.*, *20*, *1*, *29–34*, 2013.
33. Chong, A.Y.L. and Chan, F.T., Predictive analytics in precision marketing: Tools and strategies. *J. Bus. Res.*, *91*, *12–24*, 2018.
34. Voigt, P. and Von demBussche, A., The EU General Data Protection Regulation (GDPR), in: *A Practical Guide*, 1st Ed., Springer International Publishing, 2017. https://www.springerprofessional.de/en/the-eu-general-data-protection-regulation-gdpr/13934140.
35. Calif. Civ. Code § 1798.100 et seq.
36. Dastin, J., Amazon scraps secret AI recruiting tool that showed bias against women, Reuters, 2018. https://www.reuters.com/article/world/insight-amazon-scraps-secret-ai-recruiting-tool-that-showed-bias-against-women-idUSKCN1MK0AG/.
37. Vincent, J., Google 'fixed' its racist algorithm by removing gorillas from its imagelabeling tech, The Verg, 2018. https://www.theverge.com/2018/1/12/16882408/google-racist-gorillas-photo-recognition-algorithm-ai.
38. Jared Council, How adobe's ethics committee helps manage AI bias; A diverse selection of voices can help companies spot potential prob-

lems, 2021, https://www.wsj.com/articles/how-adobes-ethics-committee-helps-manage-ai-bias-11620261997.

39. Dilmegani, C., Bias in AI: What it is, Types, Examples & 6 Ways to Fix it in 2022, AIMultiple, 2022.
40. Brit, P., Tips for battling bias in ai-based personalization, 2021, https://www.destination-crm.com/Articles/Editorial/Magazine-Features/Tips-for-Battling-Bias-in-AI-Based-Personalization-147143.aspx.
41. The Conversation, Artificial Intelligence can discriminate on the basis of race, gender and also age, 2022, https://theconversation.com/artificial-intelligence-can-discriminate-on-the-basis-of-race-and-gender-and-also-age-173617.
42. Miller, A. and Hosanagar, K., Personalized discount targeting with causal machine learning, 2020.
43. Coghlan, S., Leins, K., Sheldrick, S., Cheong, M., Gooding, P., D'Alfonso, S., To chat or bot to chat: Ethical issues with using chatbots in mental health. *Digital Health*, 9, 20552076231183542, 2023.
44. Martinez-Martin, N. and Kreitmair, K., Ethical issues for direct-to-consumer digital psychotherapy apps: addressing accountability, data protection, and consent. *JMIR Ment. Health*, 5, 2, e9423, 2018.
45. Mannelli, C., Tissue vs liquid biopsies for cancer detection: ethical issues. *J. Bioeth. Inq.*, 16, 4, 551–557, 2019.
46. Denecke, K., An ethical assessment model for digital disease detection technologies. *Life Sci. Soc. Policy*, 13, 1, 16, 2017.
47. Gariépy-Saper, K. and Decarie, N., Privacy of electronic health records: a review of the literature. *J. Can. Health Libr. Assoc.*, 42, 1, 74–84, 2021 Apr 2, doi: 10.29173/jchla29496. PMID: 35949500; PMCID: PMC9327609.
48. Borgetti, S.A., Clapham, P.J., Young, J.D., Telehealth: exploring the ethical issues. *DePaul J. Health Care Law*, 19, 1, 2017.
49. Tayefi, M., Ngo, P., Chomutare, T., Dalianis, H., Salvi, E., Budrionis, A., Godtliebsen, F., Challenges and opportunities beyond structured data in analysis of electronic health records. *Wiley Interdiscip. Rev. Comput. Stat.*, 13, 6, e1549, 2021.
50. Kerwin, D., International migration and work: charting an ethical approach to the future. *J. Migr. Hum. Secur.*, 8, 2, 111–133, 2020.

51. Melé, D., Ethics at the workplace in the fourth industrial revolution: A Catholic social teaching perspective. *Bus. Ethics Environ. Responsib.*, 30, 4, 772–783, 2021.
52. McKinsey & Company, The potential for AI in the energy sector, 2020.
53. World Economic Forum, Harnessing Artificial Intelligence for the Earth, 2018.
54. European Commission, Precision farming, 2019.
55. World Wildlife Fund, Using AI to protect wildlife, 2020.
56. Strubell, E., Ganesh, A., McCallum, A., Energy and Policy Considerations for Deep Learning in NLP, 2019.
57. UNU, The Global E-Waste Monitor 2020, 2020.
58. Environmental Protection Agency, Rare Earth Elements: A Review of Production, Processing, Recycling, and Associated Environmental Issues, 2012.
59. Stapf-Fine, H., Bartosch, U., Bauberger, S., Damm, T., Engels, R., Rehbein, M., Schmiedchen, F., Sülzen, A., Policy Paper on the Asilomar Principles on Artificial Intelligence, 2018.
60. Chatila, R. and Havens, J., The IEEE Global Initiative on Ethics of Autonomous and Intelligent Systems, 2019, 10.1007/978-3-030-12524-0_2.
61. https://op.europa.eu/en/publication-detail/-/publication/d3988569-0434-11ea-8c1f-01aa75ed71a1.
62. European Commission: Directorate-General for Communications Networks, Content and Technology, *Ethics guidelines for trustworthy AI*, Publications Office, 2019, https://data.europa.eu/doi/10.2759/346720.
63. *AI in the UK: ready, willing and able?* Publisher: The Authority of the House of Lords, 2018. https://www.parliament.uk/external/committees/lords-select/ai-committee/news/2018/ai-report-published/.

Note

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